



Exoplanet 1000+

The number of discovered Extrasolar planets has risen above a thousand. What are Exoplanets? Read On: <http://dgit.in/1fUYmcn>



Telescope in the Desert

Construction of a massive telescopes in desert regions. Follow the link to see more <http://dgit.in/1bVXXc0>

The ExoMars Program currently consists of two future missions, the Orbiter and the Rover. The program as a whole is dedicated to answering the questions raised about the existence of life on Mars. The next mission in this program is the Trace Gas Orbiter.

The ExoMars TGO a project being carried forward by the ESA in collaboration with Roscosmos (The Russian Space Program) in 2016. It is a satellite which is being sent to Mars' orbit to detect the trace amounts of gases minutely detected in the atmosphere of Mars by previous missions. The measurements carried out by the Orbiter will provide evidence about the location and sources of these gases. These locations could then be targeted as possible landing sites for future missions. The Orbiter will also function as a signal booster to any future manned and unmanned rovers sent to the planet's surface. As a result of NASA's Electra technology (which is installed in the TGO) the satellite can transmit accurate Doppler data pinpointing the rover's position on Mars' surface. The TGO also uses its nadir-pointed antennae (antennae that are pointed straight down at the surface) it transmits UHF coverage to the landers and rovers on the surface which do not have adequate radio power to communicate directly with Earth.

In the wake of the anticipated success of the ExoMars Orbiter program, the ESA also have the ExoMars Rover program in the pipeline. The launch date has been tentatively set sometime in 2018. The Rover has several goals, the most important of which is exobiological and geochemistry research. The Rover is made to autonomously drill 2 metres

into the Martian crust while investigating the borehole wall mineralogy, and collect small samples. These samples will be delivered to the analytical lab inside the Rover. The sample will be crushed into a fine powder. By means of a dosing station the powder will then be presented to other instruments to perform detailed analyses of these samples. Quite the Asimovian procedure.

ISRO's advent to martian space:

Moving a little closer to home, the Indian Space Research Organization is trying to make its own headway into research on the red planet. ISRO's Mars Orbiter Mission (MOM) is scheduled for launch between 28th October and 16th November this year.

The MOM has certain technological and scientific objectives (though the former seem to outweigh the latter, probably considering how new ISRO is to the inter-planetary missions). As per ISRO's official website, they are:

Technological objectives:

- Design and realisation of a Mars orbiter with a capability to survive and perform Earth-bound manoeuvres, cruise phase of 300 days, Mars orbit insertion / capture, and on-orbit phase around Mars.
- Deep space communication, navigation, mission planning and management.
- Incorporate autonomous features to handle contingency situations.

Scientific objectives:

- Exploration of Mars surface features, morphology, mineralogy and Martian atmosphere by indigenous scientific instruments.

Have to say, it sounds very attractive in writing. But missions like the MOM are only as good as the final goal, which also hugely depends on the next big step we take towards it. However, considering we're at the eve of this launch, let's hope there's more to come. It's about time Indians began colonizing off-planet.

Dennis Tito of Mars (And other stories)

Here's hoping he at least christens the vessel Barsoom. Admit it, at least some



Dennis Tito - Space tourist

of you laughed when you heard about this guy offering a (possibly one-way) flyby trip to Mars and back, and said "Who's going to take this bozo seriously?"

Answer: At least 200,000 people. Yes, at least. The requests are still flowing in. For those who come in late: Dennis Tito is a multi-million-dollar American Engineer. He is best known for spending about twenty million dollars on a trip to the International Space Station simply to satisfy his need to see space, thus becoming the world's first Space Tourist. Now he's asking for people to join him on his trip to the red planet. He's started his own non-profit organization called Inspiration Mars and is oddly tag-lined A Mission For America. One can imagine the JLA theme song playing while he and his team walk up the ramp into the spacecraft in slow-motion, wearing spacesuits and carrying helmets under their arms.

Adding to Dennis Tito's urges and make this a mix is the Mars One project, which is offering to train a human crew to go on a one-way trip to Mars to colonize in 2023. The search for Astronauts began in April 2013 and more than 78,000 heads worldwide signed-up for the selection programme within two weeks of its launch.

*"Yes I'm an alien
When I'm touching the earth
Call me a spaceman
When I travel the universe...
...Call me a spaceman."
– Mr Spaceman, by Hardwell. *



Colonizing Mars. Science-Fiction made real?